

MediBook

Doctor Appointment Booking System

Database Systems Engineering
& Database Design

Team: 14

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Problem Statement

Challenges in traditional workflows

- Booking may rely on manual coordination
- Doctor availability can be hard to check
- Duplicate or conflicting bookings may occur
- Patient and doctor details can be scattered
- Appointment status can be difficult to track
- Access may not be separated by user role

MediBook: proposed system

- Centralize appointment-related information
- Support patient, doctor, and admin roles
- Connect the frontend to backend APIs
- Store structured data in a relational database
- Use token-based access for protected requests

Project Objectives

01

Digitize

Doctor appointment booking

02

Organize

Store core records in MySQL

03

Separate

Provide role-based access

04

Connect

Expose REST APIs

05

Coordinate

Reduce conflicting bookings

06

Protect

Use JWT token authentication

07

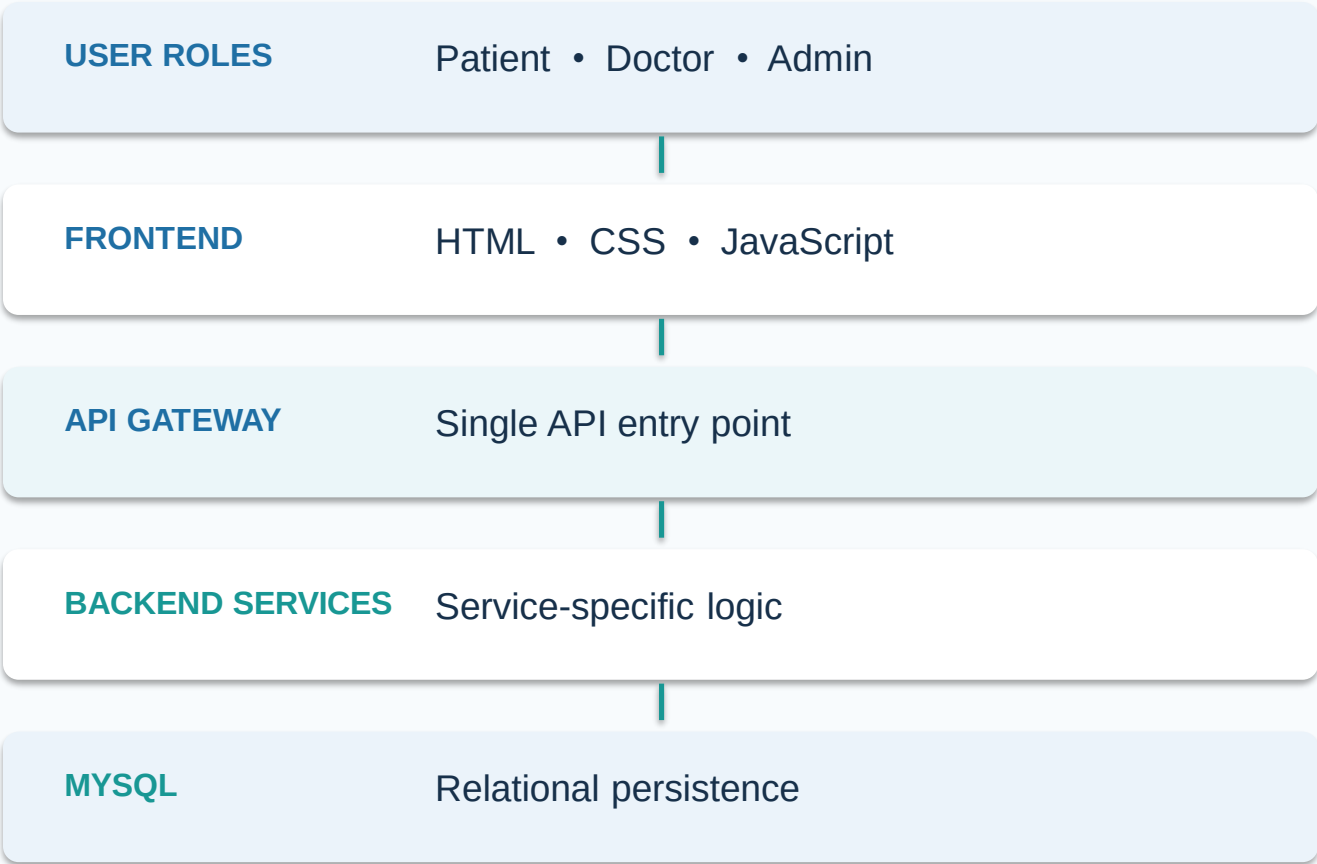
Structure

Demonstrate microservices

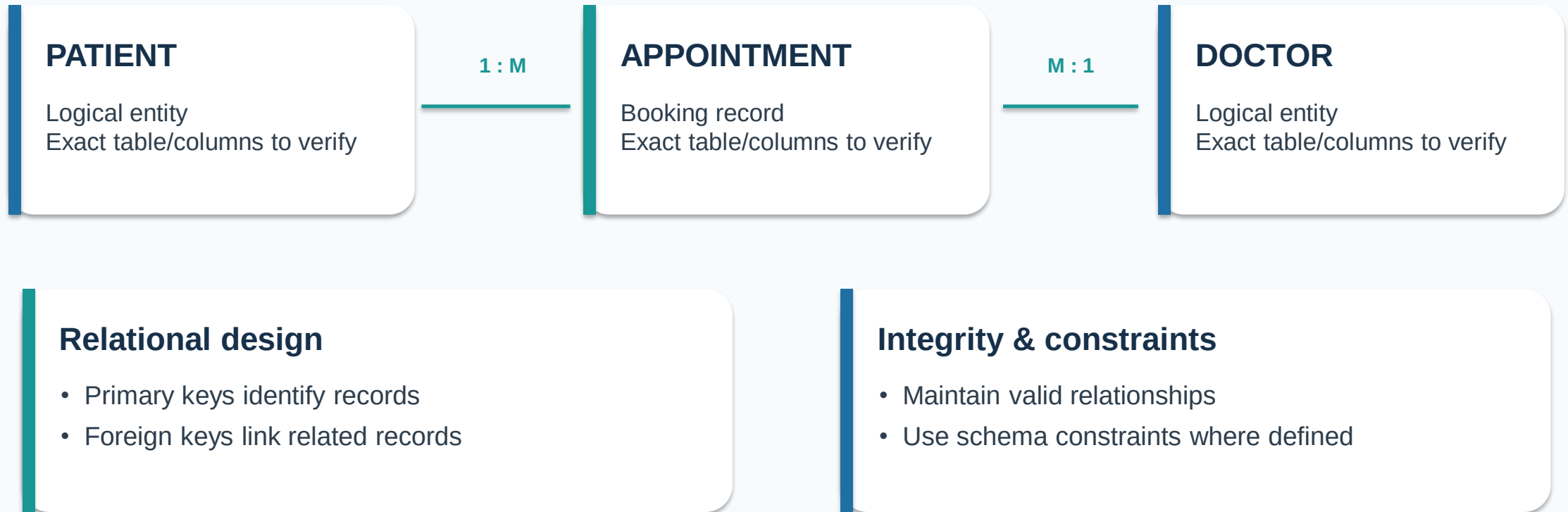
System Overview & Workflow

Request / response flow

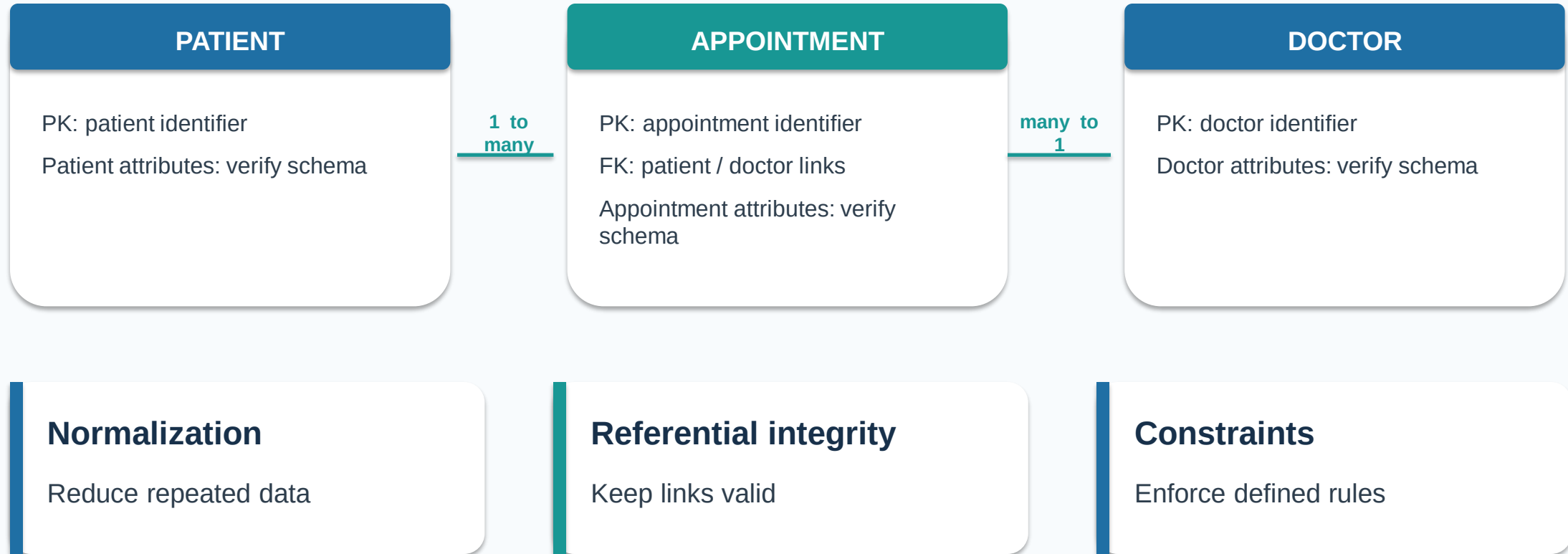
- User signs in
- Token is issued after validation
- Frontend sends protected request
- Service processes request
- Database is read or updated
- Response returns to frontend



Database Design



ER Diagram & Relational Design



Application Modules

PATIENT

- Sign in
- Find / view doctors
- Book appointment
- View / manage bookings

DOCTOR

- Sign in
- View appointments
- Appointment-related actions
- Relevant patient details

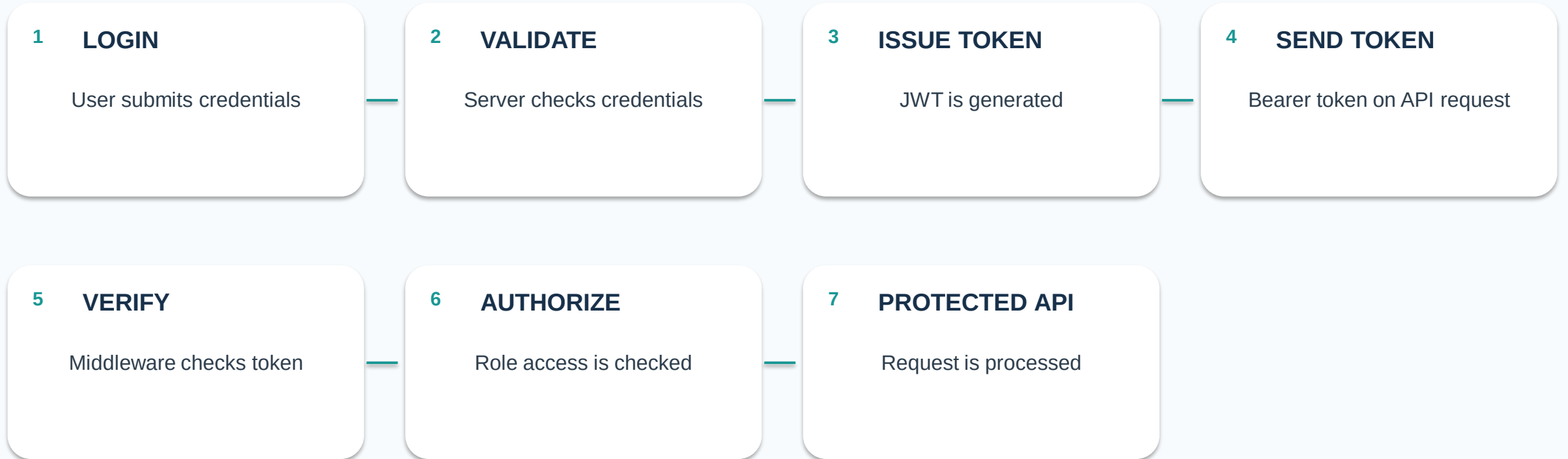
ADMIN

- Administrative access
- Patient / doctor management
- Statistics only if present

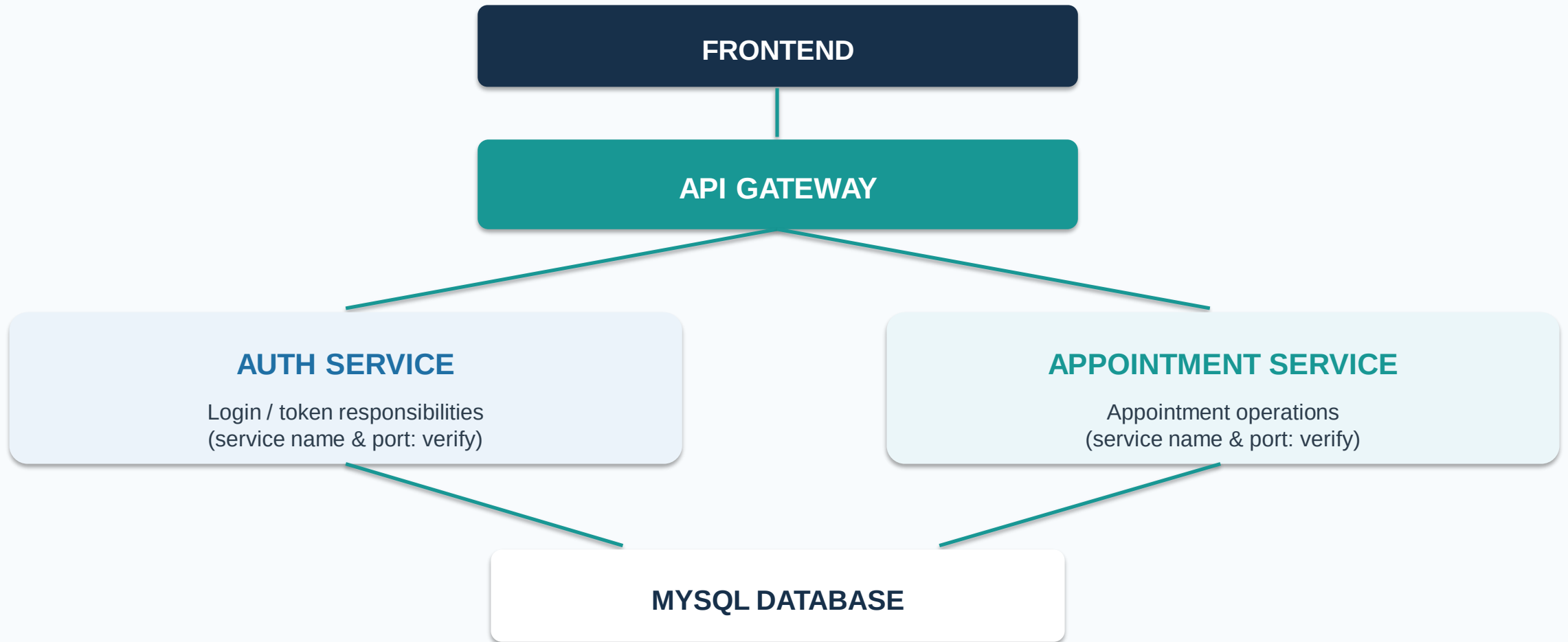
DATABASE

- Persistent records
- Relational links
- Keys & constraints

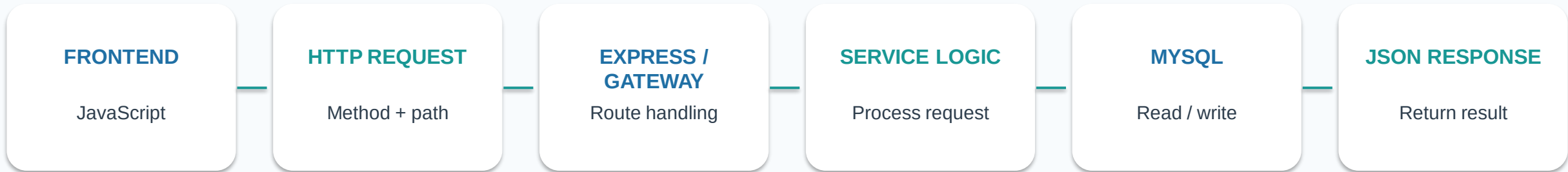
JWT Token-Based Authentication



Microservices Architecture



REST APIs & Backend



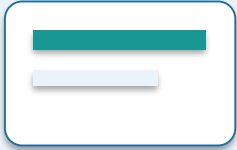
HTTP methods

- GET: retrieve data
- POST: submit or create data
- PUT / PATCH: update • DELETE: remove

Endpoint examples

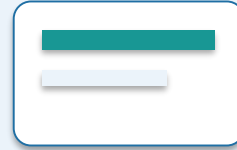
- Doctor, patient and appointment routes are in scope
- Use exact paths and methods from current route files

System Demonstration



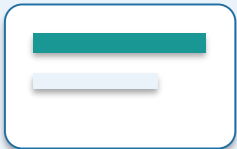
LOGIN

Insert actual login screenshot



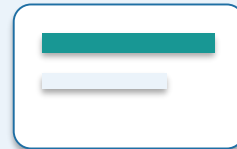
PATIENT VIEW

Insert actual patient screen



DOCTOR VIEW

Insert actual doctor screen



ADMIN / DATABASE

Insert actual admin or DB view

Syllabus Mapping & Conclusion

CONCEPT	WHERE USED IN MEDIBOOK
Database design	MySQL relational database
Primary / foreign keys	Link records and identify rows
Normalization	Organize related information
SQL	Schema, seed and setup scripts named in brief
REST APIs	Express-based backend communication
Authentication / authorization	JWT and role-based access in project brief
Microservices	Gateway and backend service structure
Frontend integration	JavaScript API communication